Detecting Fake Rumor Diffusion Through Network Visualization

Team Cheromelons

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Abstract

The rapid increase of misinformation on social media platforms presents a significant challenge to integrity of digital platforms. While traditional detection approaches focus primarily on analysing the content of the tweet, this project looks over the structural shape of rumor diffusion to understand how fake news spreads through user connections. Using a dataset consisting of **392 nodes**(Users) and **1,168 edges**(Interactions like reposts or replies) from Twitter, we constructed a directed interaction graph to model the flow of information.

We used the Gephi network visualization tool to calculate **Degree Centrality**, allowing us to identify high-influence nodes. The application of the **Fruchterman-Reingold** force-directed algorithm, the network has revealed the formation of distinct, polarized clusters. Our analysis identified specific **Super-Spreader** nodes, which were primarily Bots and Influencers that use their high connectivity to amplify fake news. Furthermore, the visualization demonstrated a critical structural isolation of **Debunkers**, effectively preventing corrective information from penetrating the rumors' **echo chambers**. These findings suggest that detecting fake news requires a network-centric approach that focuses on the connectivity of central hubs rather than content alone.

1 Introduction

1.1 Background

The arrival of Web 2.0 has shifted the pattern of information consumption from a vertical model to a horizontal model. Which means that in this network, the **Fake News** does not spread randomly, it uses the vulnerabilities in the structure of a social network.

1.2 Problem Statement

The existing countermeasures against the spread of rumors only rely on the content in the tweets. However, we have analyzed the connection structure to identify the path a rumor follows to travel from one group to another.

1.3 Objectives

The primary objectives of this study were:

1. To construct a graph network of Twitter interactions using a dataset consisting of 392 nodes.
2. To use **Gephi** to calculate the **Degree Centrality** of every node.
3. To detect the nodes which have the most connections(they control the flow of information).
4. To visualize the layout of the network to see how the fake news passes between different types of user groups.

2 Theoretical Background

To scientifically analyze the network, we relied on Graph Theory and Centrality measures.

2.1 Graph Theory Definitions

We modeled the social network as a directed graph $G = (V, E)$.

- **Vertex Set(V):** Represents the 392 distinct twitter accounts.
- **Edge Set(E):** Represents the 1168 interactions.

An edge exists from node i to node j if user i interacts with user j.

2.2 Degree Centrality

The main method used in this project to detect important nodes was degree centrality. In a directed network, it is defined as:

$$C_D(v) = \deg_{in}(v) + \deg_{out}(v) \tag{1}$$

- **In-Degree:** How many people are reacting to a user.
- **Out-Degree:** How many people the user is interacting with.

By calculating this for all 392 nodes, we were able to rank them. Nodes with a high score are "Hubs", they have the most connections and are responsible for passing news to multiple number of people irrespective of its accuracy.

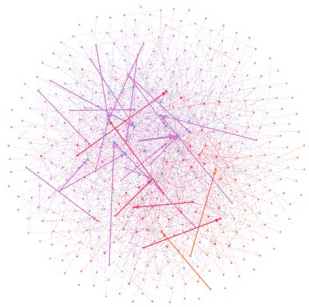

Figure 1: Graph of the data after applying Fuchterman-Reingold Algorithm

2.3 The Fruchterman-Reingold Algorithm

To reveal the clusters in this data, we imported it to Gephi and applied the Fruchterman-Reingold Force-Directed Layout. This specific algorithm was chosen as it treats edges like springs, it pulls the connected nodes together and pushes the disconnected nodes further apart. This was essential to find out the communities formed based on the frequency of the users' interaction.

3 Methodology

We used an open source dataset "PHEME" from **Kaggle**. The dataset contains user IDs and interaction data suitable for network analysis.

Table 1: Dataset Statistics

Parameter	Value	Description
Total Nodes (V)	392	Distinct User Accounts
Total Edges (E)	1,168	Interactions (Retweets/Replies)
Avg. Degree	2.98	Avg. interactions per user
Graph Density	0.0076	Sparse network structure
Node Categories	4	Naive, Bot, Debunker, Influencer

3.1 Graph Construction

The detection process followed these main steps:

1. **Import:** Loaded the network of 392 nodes and 1168 edges.
2. **Coloring:** We assigned colors to each type of users.

3. **Centrality Calculation:** We ran the "Average Degree"statistic in Gephi to calculate the connection count of every node.

4. **Sizing:** We set the node size to be proportional to iy=ts Degree Centrality. Nodes with more connections were made larger (Mun size:10, Max size:60).

5. **Layout:** Applied Fruchterman-Reingold to space out the nodes.

6. **Filtering:** We examined the largest nodes to see their category (Bots vs Human).

3.2 Visualization Layout

We sorted our nodes into four categories:

Table 2: User Roles

User Category	Percentage	Color	Role Description
Naive User	71.75%	Purple	Regular users.
Bot	13.75%	Red	Automated accounts.
Debunker	10.25%	Orange	Fact-checkers.
Influencer	0.25%	Blue	High-reach accounts.

4 Results

After applying the layout and filtering the data, three distinct structures have emerged.

4.1 Polarization and Echo Chambers

The network did not form a single unified structure. Instead, it has separated into distinct structures with high modularity scores each.

Observation: The "Rumor Cluster" (represented in red/pink in the visualization) formed a tight, isolated chamber.

Interpretation: Rumor supporters interact almost exclusively with other supporters. This creates a bubble where the misinformation thrives without any outside (debunkers) interference.

4.2 Identification of High Degree Nodes

By scaling a nodes size based on its degree centrality, we have observed that a very small number of nodes appear significantly larger than the rest. These "Super Spreaders" were primarily identified to be either bots or influencers.

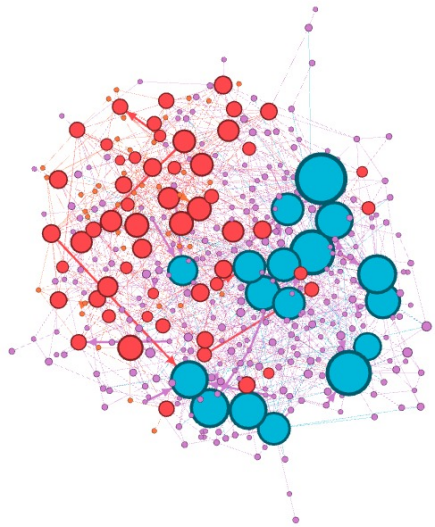

Figure 2: Final graph

Obervation: These super spreaders make up approximately 14% of the network and commanded the majority of the connections with their influence.

Interpretation: A small minority of accounts are responsible for massively accelerating the spread of the rumor. They act as the bridges that push content to a large mass of the naive users.

4.3 Weak Cross-Boundary Flow

Perhaps the most important finding from this was the lack of edges from the Rumor cluster to the debunker cluster.

Observation: Edges from the debunker cluster rarely entered the rumor cluster.

Interpretation: Fact checks made by the debunkers mostly circulate within the debunker group itself. Structurally, there are very few ways for a fact-check to reach the users inside the rumor's echo chamber.

5 Analysis

Our visualization clearly shows that the misinformation spread is a structural problem. The Fruchterman-Reingold layout revealed that the network is in pieces with very weak links between each piece.

Centrality measurements highlight that the high connectivity of bots creates an amplification of the rumor. This is because the naive users are closer to

the bots and influencers in the graph, who spread the rumors without proper verification.

On the other hand, the Debunkers suffer from structural isolation. Even if their content is factual, the network topology prevents it from flowing into the rumor cluster. This is why debunking efforts often fail, the "truth" eventually hits a wall due to the lack of edges between the clusters.

6 Conclusion

This project demonstrates that combining fake news requires more than just content moderation or fact checking. Our analysis has led us to three main conclusions:

- **Structure over Content:** Diffusion is strongly driven by the topology of the network.
- **The Bot Factor:** A small percentage of highly connected bots boosts the rate of rumor spread exponentially.
- **The Reach Problem:** Rumor networks form impenetrable clusters.

Since the debunkers are locked out of the rumor chambers, we conclude that intervention strategies must target the "Central Nodes" or Super Spreaders rather than attempting to flood the network with passive fact-checks.